1. Write a C program to find the maximum of three numbers without using logical operators.

main.c

```c
#include <stdio.h>

int main() {
    int a, b, c, max;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a > b) {
        if (a > c)
            max = a;
        else
            max = c;
    } else {
        if (b > c)
            max = b;
        else
            max = c;
    }
    printf("Maximum number is: %d", max);

    return 0;
}
```

Output

```
Enter three numbers: 45
50
3
Maximum number is: 50

=== Code Execution Successful ===
```

2. Write a C program to check whether the given year is leap year or not by adding century leap year or non-century leap year in the output (Eg: 2000 is a century leap year, 2024 is a non-century leap year)

main.c

```c
#include <stdio.h>

int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if (year % 400 == 0) {
        printf("%d is a century leap year", year);
    }
    else if (year % 100 == 0) {
        printf("%d is not a leap year", year);
    }
    else if (year % 4 == 0) {
        printf("%d is a non-century leap year", year);
    }
    else {
        printf("%d is not a leap year", year);
    }

    return 0;
}
```

Output

```
Enter a year: 2026
2026 is not a leap year

=== Code Execution Successful ===
```

3. Write a C program to find whether the entered character is alphabet / digit / special character. If the entered character is an alphabet then say it is vowel or consonant without using built in functions.

main.c

```c
#include <stdio.h>

int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch);
    if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
        printf("It is an alphabet\n");
        if (ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U' ||
            ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
            printf("It is a vowel");
        } else {
            printf("It is a consonant");
        }
    }
    else if (ch >= '0' && ch <= '9') {
        printf("It is a digit");
    }
    else {
        printf("It is a special character");
    }

    return 0;
}
```

Output

```
Enter a character: apple
It is an alphabet
It is a vowel

=== Code Execution Successful ===
```

4. Write a C program for simple ATM simulation with operations Check Balance, Deposit, Withdraw, Exit using switch and update balance accordingly.

main.c

```c
#include <stdio.h>

int main() {
    int choice;
    float balance = 1000.0, amount;

    do {
        printf("\n--- ATM MENU ---\n");
        printf("1. Check Balance\n");
        printf("2. Deposit\n");
        printf("3. Withdraw\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Current Balance: %.2f\n", balance);
                break;

            case 2:
                printf("Enter amount to deposit: ");
                scanf("%f", &amount);
                if (amount > 0) {
                    balance += amount;
                    printf("Amount deposited successfully.\n");
                } else {
                    printf("Invalid amount.\n");
                }
                break;

            case 3:
                printf("Enter amount to withdraw: ");
                scanf("%f", &amount);
                if (amount > 0 && amount <= balance) {
                    balance -= amount;
                    printf("Withdrawal successful.\n");
                } else {
                    printf("Insufficient balance or invalid amount.\n");
                }
                break;

            case 4:
                printf("Thank you for using ATM.\n");
                break;

            default:
                printf("Invalid choice. Try again.\n");
        }

    } while (choice != 4);

    return 0;
}
```

Output

```
--- ATM MENU ---
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 200
Invalid choice. Try again.

--- ATM MENU ---
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 1
Current Balance: 1000.00

--- ATM MENU ---
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice: 2
Enter amount to deposit: 1000
Amount deposited successfully.

--- ATM MENU ---
1. Check Balance
2. Deposit
3. Withdraw
4. Exit
Enter your choice:
```

5. Write a C program for menu driven calculator using switch statement.

main.c

Share Run

```c
#include <stdio.h>

int main() {
    int choice;
    float a, b, result;

    do {
        printf("\n--- CALCULATOR MENU ---\n");
        printf("1. Addition\n");
        printf("2. Subtraction\n");
        printf("3. Multiplication\n");
        printf("4. Division\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                result = a + b;
                printf("Result = %.2f\n", result);
                break;

            case 2:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                result = a - b;
                printf("Result = %.2f\n", result);
                break;

            case 3:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                result = a * b;
                printf("Result = %.2f\n", result);
                break;

            case 4:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                if (b != 0) {
                    result = a / b;
                    printf("Result = %.2f\n", result);
                } else {
                    printf("Error! Division by zero.\n");
                }
                break;

            case 5:
                printf("Exiting calculator...\n");
                break;

            default:
                printf("Invalid choice! Try again.\n");
        }

    } while (choice != 5);

    return 0;
}
```

Output

```
--- CALCULATOR MENU ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice: 2
Enter two numbers: 23
03
Result = 20.00

--- CALCULATOR MENU ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice:
```